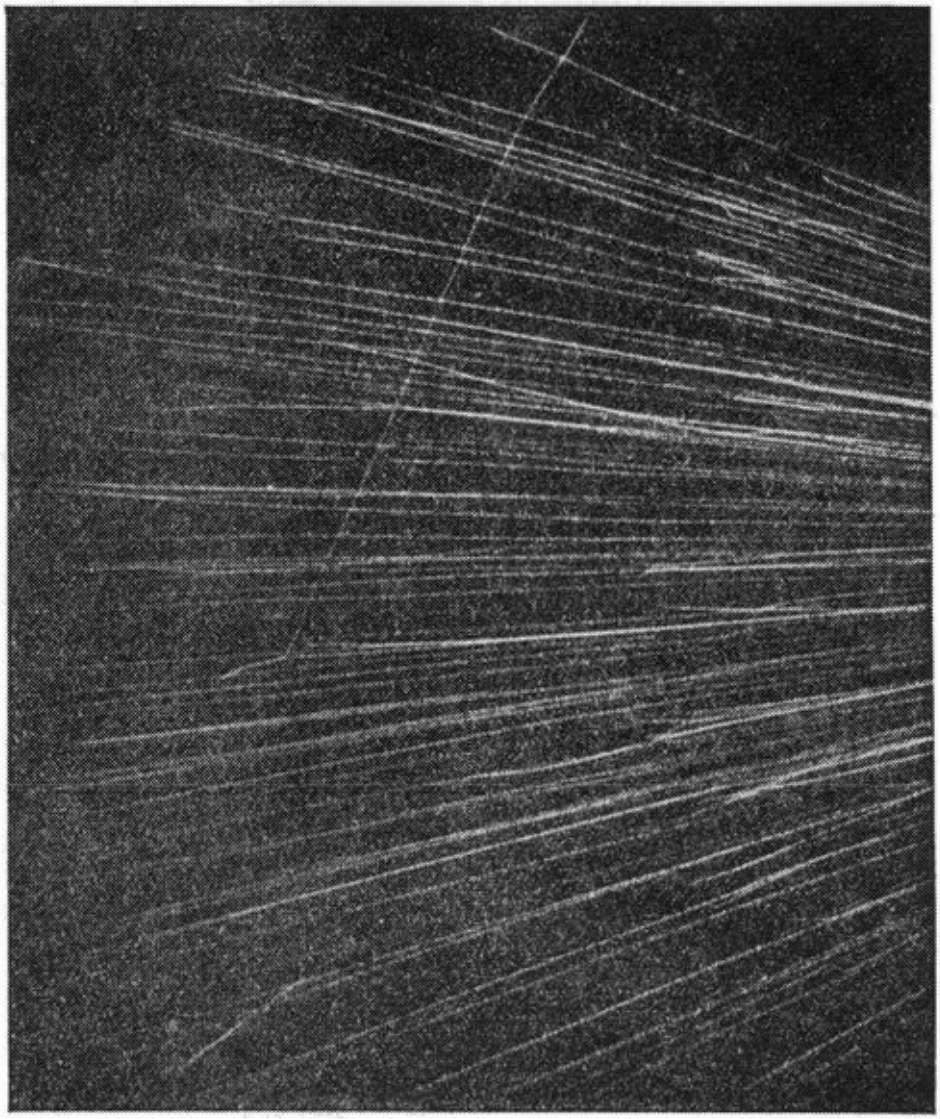


Figure 10.—Transformation of a nitrogen nucleus into an oxygen nucleus.

As we see, in this process, a particle of mass number 1 and charge (atomic) number 0 is ejected—a neutron. We designate this particle by the symbol ${}_0n^1$, and we can express this *nuclear reaction* (this is the name given to such processes) as well as those previously discussed, by formulae. We take into consideration also that in addition to the neutron a gamma-ray photon (its symbol is y) is frequently emitted, too. The formula of the nuclear reaction just mentioned reads, therefore:

$$ {}_4Be^9 + {}_2He^4 \rightarrow {}_6C^{12} + {}_0n^1 + \gamma $$